

# What Predicts a Movie's Financial Success?

*DS2500 – Team Project Final Report*

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## 1. Introduction

### **Goal of the project:**

This project dives into whether a movie's features (budget, runtime, and genre) can help us predict whether it earns its budget back, and separately, how much revenue it is likely to make.

### **Problem Statement and Background:**

Making a movie is a huge financial gamble. The cost of creating a single film/movie averages around \$50 million to \$200 million, and a large portion of movies usually never make that money back. So they have to be cautious when making decisions without knowing whether the movie will succeed or not, so there is a big decision-making process that comes down to intuition and experience rather than evidence.

Narrowing the problem down into a question we can test using only information we would know before the release of the movie, we have three main factors: budget, runtime, and genre. Can we predict whether a movie will make its money back (revenue

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greater than budget, meaning it's a hit)? Answering this can help us determine a movie's financial result from basic features.

## Why This Matters

- The problem is, like I said before, studios take financial gambles every time they produce a movie. Knowing the basic features that line up with the risk can help determine a movie's risk.
- For anyone studying or in this business, this gives a measured answer to how much a movie's financial gamble comes down to.

## 2. Dataset

### Dataset Overview

Source: TMDB 5000 Movie Dataset (<https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata>), published on Kaggle.

Time period: The dataset covers movies back to early 1900 and up to 2017 which is when this dataset was updated

Size: 4803 movies with 20 columns. After cleaning the data based on missing data (0), it went down to 3226 movies left

Format: CSV file

### Data collection methodology

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TMDB pulled it together from studio-provided information and from a group of volunteer editors, so it reflects information those contributors happened to know for each movie. It isn't a carefully sampled dataset of every movie ever made.

## Key Variables

| Variable      | Description                                            | Type        | Example                   |
|---------------|--------------------------------------------------------|-------------|---------------------------|
| budget        | Reported production budget in USD                      | Numeric     | 12345678                  |
| revenue       | Total box office gross worldwide in USD                | Numeric     | 9876543                   |
| runtime       | length in minutes                                      | Numeric     | 135                       |
| genres        | list of genre tags                                     | Text        | [{"name": "Action"}, ...] |
| primary_genre | First genre listed, used as a simplified genre feature | Categorical | Action                    |
| hit_or_flop   | 1 if revenue > budget, else 0                          | Binary      | 1                         |

## Ethical Considerations

- Privacy: this dataset is about movies and companies, not people, so there isn't much personal privacy risk
- Reporting bias; budget and revenue numbers are self-reported and often missing for moves that are independent or older. Removing those rows pushes data towards better significance

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## 3. Methods

### **Data preprocessing/preparation**

Many columns had the value of zero, meaning that the information was never reported (false zero). A movie can't have a runtime of zero minutes, so I cleaned up rows where either budget, revenue, or runtime was zero, bringing down the dataset to 3226 movies.

Budget and revenue were log-transformed to fix the skewness since a small number of movies are worth more than a typical movie. With genre, I only used the first genre listed as simplification which I treated as a limitation.

The target marks a movie as a hit if its revenue was greater than or equal to its budget, otherwise a flop. This target showed an imbalance of 76% of movies being hits while the 24% are flops.

### **EDA**

Exploration of the data confirmed that revenue and budget are skewed, said previously. And budget and revenue show a positive but decent relationship, higher budget movies tend to earn more, but there's a lot of variation at every budget level.

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## Modeling & Evaluation

I used three methods. Two predict hit or flop, and one predicts revenue directly.

K-Nearest Neighbors looks at a movie's most similar past movies and guesses the outcome. Logistic Regression works as it fits one equation across the dataset instead of comparing to nearby movies

We split the data into training , validation , and test sets. For KNN, we used the validation set to pick the best number of neighbors (k). We tested odd values from 3-31 and picked the best F1 score on flops, not the best accuracy, since accuracy can be misleading. Logistic Regression used that same test set and features.

Separately, Linear Regression predicted revenue from the same three features. I reported  $R^2$  and mean absolute error as the metrics.

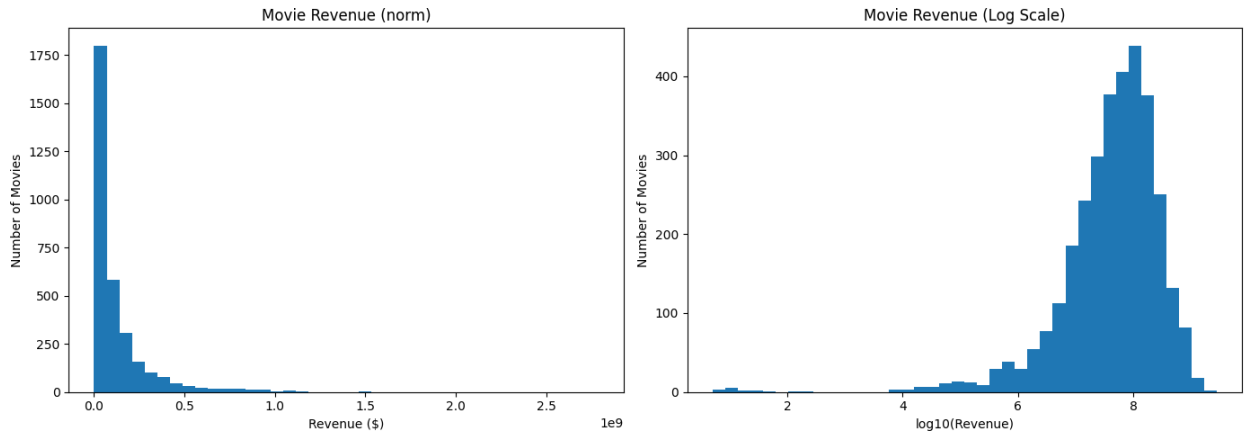
## 4. Results and Conclusions

### **Finding 1: Revenue and budget are both skewed.**

A small number of movies earn far more than a typical movie in this dataset. That's why both budget and revenue were log transformed before any further analysis. This is a data preparation step, but it also means financial success isn't spread out evenly. A small number of hits bring in a disproportionate share of all the revenue

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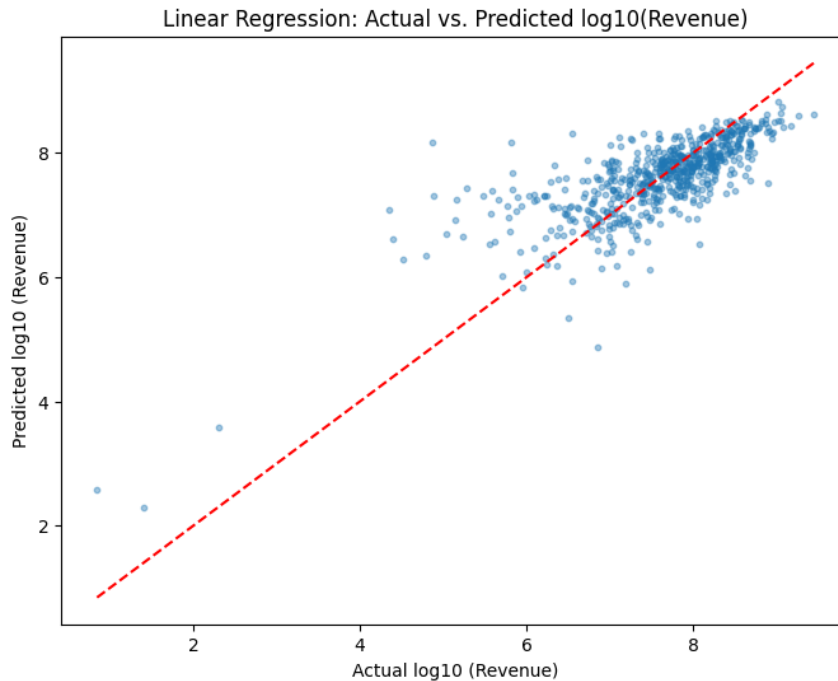
### Finding 2: Don't trust the accurate model! Uh ohhh

Logistic Regression had higher accuracy than KNN (75.7% vs. 71.8%), but its F1 score on the flop class, which is the metric I used to interpret these models since flops are the harder and more useful outcome to predict, was exactly 0. Logistic Regression never correctly identified a flop in the set. It just predicted the majority “hit” outcome for almost every movie. KNN actually identified some real flops due to  $F1 = 0.18$ . This is an example of why accuracy can be a misleading metric when the data is imbalanced, and why I based F1 by our models with flop.

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**Finding 3: Budget, runtime, and genre explain about 58% of the variation in revenue.**



The linear regression model, predicting log transformed revenue from budget, runtime, and genre, got an  $R^2$  of 0.577 on the test data, with a mean absolute error of 0.42 in  $\log_{10}$  units. The figure shows this directly. Predicted and actual revenue trend positive, but movies scatter around the line of best fit. If the features explain about half of what drives revenue, then no classifier built from the features could do better at predicting hit or flop.

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| Metric    | Value | Interpretation                                  |
|-----------|-------|-------------------------------------------------|
| Accuracy  | 71.8% | Overall correct-guess rate on the test set      |
| Precision | 31%   | Of movies predicted as flops, 31% actually were |
| Recall    | 13%   | Only 13% of real flops were correctly caught    |
| F1 Score  | 18%   | Low ability to identify flop                    |

## Conclusions

So can budget, runtime, and genre predict a movie's financial success, my analysis somewhat answers this. These features are measurable but they're not that reliable to predict any movie's hit or flop outcome. Our KNN classifier only caught 13% of flops, and a logisitc reg. model chosen for accuracy caught none at all. The takeaway is that movie features are a meaningful but incomplete prediction of financial risk. A better risk assessment needs greater information than just budget, runtime, and genre.

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## 5. Future work and limitations

### Limitations of Current Analysis

- Data constraints: about a third of the dataset was removed because of missing numbers, which skews the sample.
- Methodological limitation: using only the first listed genre is a limitation however limited the information for movies that had more than one genre
- Assumptions: I defined a hit as revenue at or above budget, which ignores marketing, distribution costs, and other revenue that affects a movie's real profitability.

### Future Research Directions

- Add better features like cast and crew popularity, and release timing, to see if they improve on the 58% of revenue variation.
- Use all listed genres per movie instead of just the first and add more recent movies to dataset instead of going up to 2017

## References

TMDB 5000 Movie Dataset. (2017). The Movie Database (TMDB), via Kaggle.

<https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata>